

HARDEEP SINGH

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PROFILE

Self-taught AI engineer with 200+ organic users across three deployed products and 44+ GitHub stars tools within a month. Builds and ships end-to-end LLM-powered products: Python, FastAPI, Next.js, React, and the full AI toolchain including RAG pipelines, vector search, and multi-provider LLM integration. Currently deepening RAG architecture knowledge through active project work. Proven ability to move fast under pressure, having co-built a crisis-response platform serving 15 Indian states in 48 hours.

PROJECTS

Coffee Coach

2024 – Present

Python | FastAPI | Next.js | React | PostgreSQL | Docker | LLM coaching | coffeecoach.app

- Mobile-first web app for specialty coffee drinkers to log brews, track improvement over time, and receive personalised coaching from an AI coach called Kapi.
- Acquired 64 real users organically through Twitter/X and Reddit with no paid promotion.
- Full-stack deployment: Next.js on Vercel, FastAPI on Hetzner VPS with Docker, Nginx, and Certbot. Demonstrates end-to-end product ownership from concept to production.

YouTube RAG Scraper

2026 – Present

Python | LangChain | FAISS | RAG | github.com/rav4nn/youtube-rag-scraper

- Pipeline that scrapes YouTube videos, extracts transcripts, embeds them into a FAISS vector store, and exposes a semantic search interface over the resulting knowledge base.
- 44 GitHub stars and 11 forks gained organically, the most-starred project in the portfolio.
- End-to-end RAG architecture: ingestion, chunking, embedding, retrieval, and generation.

Flux RAG

2026 – Present

Python | RAG | Vector Search | Embeddings | github.com/rav4nn/flux-rag

- Deep-dive RAG project focused on advanced retrieval architecture: chunking strategies, re-ranking, hybrid search, and retrieval evaluation.
- Built to develop rigorous understanding of the full RAG stack beyond surface-level implementation, directly applicable to production AI systems.

Splitwala — WhatsApp Expense Splitter

2026 – Present

Node.js | whatsapp-web.js | github.com/rav4nn/splitwala-webjs

- WhatsApp bot for splitting expenses across group chats, supporting commands including /split, /summary, and balance tracking.
- Used by hundreds of WhatsApp users across the world, lives on a real number
- Node.js stack broadens portfolio beyond Python and shows cross-language comfort.

Grandma-Proof — AI UX Testing Agent

2026 – Present

Python | LLM Agents | Prompt Engineering | github.com/rav4nn/grandma-proof

- AI agent that simulates a confused first-time user to stress-test web UX before real users hit friction.
- Uses multi-step reasoning to surface unintuitive UI flows, a novel application of LLM agents to product testing.

EXPERIENCE

Operations Lead — CovidWin

Apr 2021 – May 2021

Volunteer Crisis Platform | covidwin.in | Python, Google Sheets API, Cron Automation

- Led volunteer onboarding across 8 states as Operations Lead, coordinating a network of 4,000 volunteers from partner organisations U&I and Yuva Halla Bol.
- Contributed to a 15-minute automated data sync pipeline with deduplication, Google Sheets API bulk import, and multi-source aggregation.
- Platform reached full operations in 48 hours, covering 15 states and 50 cities, and verified 10,000 life-saving COVID resources during India's second wave.

Freelance Digital Marketing Consultant

2018 – 2021

SEO | Content Strategy | Social Media | Campaign Management

- Delivered SEO, content planning, and social media campaign management for small business clients over three years.
- Managed end-to-end client relationships independently.

SKILLS

Languages Python, JavaScript (Node.js), HTML/CSS, C++

AI / ML LangChain, FAISS, RAG pipelines, Prompt Engineering, Embeddings, Vector Search

LLM APIs Claude (Anthropic), OpenAI, DeepSeek, Gemini

Backend FastAPI, REST APIs, PostgreSQL, Next.js, React, TailwindCSS, Cron Jobs, Google Sheets API, whatsapp-web.js

Deployment Hetzner VPS, Docker, Nginx, Certbot, Vercel, Railway, Fly.io, Git/GitHub

EDUCATION

Diploma in Chemical Engineering

2018

Indian Institute of Technology (IIT), Delhi | CGPA: 6.535 / 10

Class XII — CBSE

2013

St. Mary's School, Delhi | 93.25% | Mathematics, Computer Science, Physics, Chemistry

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